

CAMPENI, Romania

WMO: 151620

Lat: 46.364N

Lon: 23.040E

Elev: 592

StdP: 94.41

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.6	(c) -12.1	(d) -16.5	(e) 0.9	(f) -13.4	(g) -14.2	(h) 1.2	(i) -11.0	(j) 6.8	(k) -3.1	(l) 4.2	(m) -0.3	(n) 0.5	(o) 230	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.3	(c) 30.1	(d) 20.2	(e) 28.4	(f) 19.5	(g) 26.6	(h) 18.8	(i) 21.3	(j) 28.1	(k) 20.3	(l) 26.7	(m) 19.6	(n) 25.4	(o) 1.5	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.9	23.9	18.2	14.1	23.1	17.4	13.4	22.2	64.6	28.4	61.2	26.6	58.5	25.5	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.3	(b) 3.4	(c) 3.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 4.3	(b) 3.4	(c) 3.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-2.2	0.6	4.2	8.8	13.6	17.6	18.6	19.1	14.1	9.5	3.8	-0.5
	DBStd	8.27	4.54	3.85	3.97	4.42	2.99	3.28	2.67	2.89	3.75	3.80	4.13	3.96
	HDD10.0	1484	379	264	182	74	7	0	0	0	7	56	188	327
	HDD18.3	3542	638	498	438	287	146	53	30	27	132	273	436	585
	CDD10.0	1106	0	0	2	37	119	227	266	281	131	41	2	0
	CDD18.3	123	0	0	0	0	1	29	37	50	6	0	0	0
	CDH23.3	1418	0	0	0	15	53	303	416	524	99	9	0	0
	CDH26.7	381	0	0	0	0	5	72	113	173	17	0	0	0
Wind	WSAvg	0.8	0.6	0.9	0.9	1.1	1.0	1.0	0.8	0.8	0.7	0.7	0.6	0.8
Precipitation	PrecAvg	809	42	37	48	68	86	101	107	76	72	60	55	60
	PrecMax	1045	88	104	133	129	156	180	179	191	158	142	99	137
	PrecMin	562	13	7	12	9	47	34	30	18	11	3	0	5
	PrecStd	138	21	22	32	27	30	37	41	33	37	37	24	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	13.5	20.6	25.3	27.1	30.7	31.8	32.9	29.1	24.3	16.3	9.2
		MCWB	5.5	8.7	11.5	14.9	16.7	21.5	20.4	20.7	18.0	14.4	11.5	6.7
	2%	DB	6.1	10.6	17.0	23.0	24.9	28.7	29.5	30.4	26.2	21.8	13.1	7.0
		MCWB	4.2	6.4	10.1	13.6	16.7	20.2	20.3	19.8	17.6	13.9	9.7	5.6
	5%	DB	4.4	8.0	13.6	20.5	22.8	26.9	27.5	28.5	24.0	19.1	11.1	5.4
		MCWB	3.1	4.9	8.3	12.2	15.6	19.5	19.7	19.3	16.5	13.1	8.9	4.4
	10%	DB	2.9	5.7	10.8	17.6	20.6	24.7	25.6	26.1	21.3	16.0	9.3	3.9
		MCWB	2.1	3.7	6.5	10.9	14.7	18.3	18.9	18.9	15.8	11.9	7.9	3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	9.1	12.5	15.7	18.6	22.2	22.7	22.0	19.4	15.9	12.2	7.5
		MCDB	7.9	13.7	19.9	23.8	25.5	29.7	28.3	30.1	27.2	21.6	14.9	8.2
	2%	WB	4.5	7.1	10.4	14.3	17.1	20.9	21.2	21.0	18.2	14.6	10.6	5.7
		MCDB	5.5	9.9	16.2	21.7	23.2	27.5	27.4	28.5	24.8	20.5	12.7	6.7
	5%	WB	3.3	5.3	8.8	12.9	16.1	19.9	20.2	20.1	17.1	13.4	9.4	4.3
		MCDB	4.3	7.2	13.2	19.8	21.4	25.9	26.3	26.3	22.4	18.2	10.9	5.2
	10%	WB	2.2	3.9	7.2	11.5	15.1	18.9	19.4	19.3	16.1	12.1	8.1	3.2
		MCDB	2.9	5.3	10.3	17.0	19.9	24.0	24.8	25.3	20.7	15.4	9.2	3.8
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.5	11.2	13.0	12.2	12.6	13.3	13.6	11.8	12.6	8.8	6.6
		MCDBR	8.7	13.4	17.2	18.7	16.2	15.7	16.4	17.2	16.2	17.7	12.0	7.8
		MCWBR	6.6	8.8	10.1	9.6	8.3	7.6	7.8	7.2	7.5	10.0	8.0	6.0
	5% WB	MCDBR	7.7	11.5	15.9	17.2	14.5	14.8	14.7	15.4	14.4	16.2	10.2	7.5
		MCWBR	5.9	7.9	9.6	8.9	7.7	7.4	7.5	7.0	7.0	9.2	7.2	5.8
Clear-Sky Solar Irradiance	taub	0.312	0.342	0.388	0.442	0.427	0.420	0.439	0.446	0.406	0.369	0.340	0.314	
	taud	2.316	2.249	2.189	2.074	2.180	2.238	2.192	2.168	2.249	2.338	2.363	2.352	
	Ebn at Noon	770	816	827	812	841	848	825	801	801	777	731	725	
	Edh at Noon	77	101	124	153	142	135	140	138	116	92	74	67	
All-Sky Solar Radiation	RadAvg	1.22	2.02	3.16	4.33	5.21	5.78	5.66	5.22	3.71	2.56	1.47	0.97	
	RadStd	0.17	0.26	0.46	0.46	0.45	0.49	0.50	0.50	0.42	0.30	0.20	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)		+0.71	N/A	N/A	+1.00	N/A	N/A	-125	-189	+121	+60

Nomenclature: See separate page